1.02 Algebra and Functions				
1.02a	Indices	a) Understand and be able to use the laws of indices for all rational exponents. *Includes negative and zero indices.* *Problems may involve the application of more than one of the following laws:* $x^a \times x^b = x^{a+b}, x^a \div x^b = x^{a-b}, (x^a)^b = x^{ab}$ $x^{-a} = \frac{1}{x^a}, \quad x^{\frac{m}{n}} = \sqrt[n]{x^m}, \quad x^0 = 1.$		MB1
1.02b	Surds	b) Be able to use and manipulate surds, including rationalising the denominator. *Learners should understand and use the equivalence of surd and index notation.*		MB2
1.02c	Simultaneous equations	c) Be able to solve simultaneous equations in two variables by elimination and by substitution, including one linear and one quadratic equation. *The equations may contain brackets and/or fractions.* *e.g.* $y = 4 - 3x$ *and* $y = x^2 + 2x - 2$ $2xy + y^2 = 4$ *and* $2x + 3y = 9$		MB4

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1.02d 1.02e 1.02f	Quadratic functions	d) Be able to work with quadratic functions and their graphs, and the discriminant (*D* or Δ) of a quadratic function, including the conditions for real and repeated roots. *i.e. Use the conditions:* *1.* $b^2 - 4ac > 0 \Rightarrow$ *real distinct roots* *2.* $b^2 - 4ac = 0 \Rightarrow$ *repeated roots* *3.* $b^2 - 4ac < 0 \Rightarrow$ *roots are not real* *to determine the number and nature of the roots of a quadratic equation and relate the results to a graph of the quadratic function.* e) Be able to complete the square of the quadratic polynomial $ax^2 + bx + c$. *e.g. Writing* $y = ax^2 + bx + c$ *in the form* $y = a(x+p)^2 + q$ *in order to find the line of symmetry* $x = -p$*, the turning point* $(-p,\ q)$ *and to determine the nature of the roots of the equation* $ax^2 + bx + c = 0$ *for example* $2(x+3)^2 + 4 = 0$ *has no real roots because* $4 > 0$. f) Be able to solve quadratic equations including quadratic equations in a function of the unknown. *e.g.* $x^4 - 5x^2 + 6 = 0$, $x^{\frac{2}{3}} - 5x^{\frac{1}{3}} + 4 = 0$ *or* $\dfrac{5}{(2x-1)^2} - \dfrac{10}{2x-1} = 1$.		MB3

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1.02g	Inequalities	g) Be able to solve linear and quadratic inequalities in a single variable and interpret such inequalities graphically, including inequalities with brackets and fractions. *e.g.* $10 < 3x + 1 < 16$, $(2x+5)(x+3) > 0$. [*Quadratic equations with complex roots are excluded*.]		MB5
1.02h		h) Be able to express solutions through correct use of 'and' and 'or', or through set notation. *Familiarity is expected with the correct use of set notation for intervals, e.g.* $\{x : x > 3\}$, $\{x : -2 \le x \le 4\}$, $\{x : x > 3\} \cup \{x : -2 \le x \le 4\}$, $\{x : x > 3\} \cap \{x : -2 \le x \le 4\}$, $\varnothing$. *Familiarity is expected with interval notation, e.g.* $(2, 3)$, $[2, 3)$ *and* $[2, \infty)$.		
1.02i		i) Be able to represent linear and quadratic inequalities such as $y > x + 1$ and $y > ax^2 + bx + c$ graphically.		

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1.02j 1.02k	Polynomials	j) Be able to manipulate polynomials algebraically. *Includes expanding brackets, collecting like terms, factorising, simple algebraic division and use of the factor theorem.* *Learners should be familiar with the terms "quadratic", "cubic" and "parabola".* *Learners should be familiar with the factor theorem as:* *1.* $\mathrm{f}(a) = 0 \Leftrightarrow (x - a)$ *is a factor of* $\mathrm{f}(x)$*;* *2.* $\mathrm{f}(\frac{b}{a}) = 0 \Leftrightarrow (ax - b)$ *is a factor of* $\mathrm{f}(x)$*.* *They should be able to use the factor theorem to find a linear factor of a polynomial normally of degree* ≤ 3*. They may also be required to find factors of a polynomial, using any valid method, e.g. by inspection.*	k) Be able to simplify rational expressions. *Includes factorising and cancelling, and algebraic division by linear expressions.* *e.g. Rational expressions may be of the form* $\frac{x^3 - x - 2}{2x + 1}$ *or* $\frac{(x^2 - x - 6)(x^2 + 4x + 3)}{(x^2 - 9)(x + 3)}$. *Learners should be able to divide a polynomial of degree* ≥ 2 *by a linear polynomial of the form* $(ax - b)$*, identify the quotient and remainder and solve equations of degree* ≤ 4*.* *The use of the factor theorem and algebraic division may be required.*	MB6
1.02l	The modulus function		l) Understand and be able to use the modulus function, including the notation $\lvert x \rvert$, and use relations such as $\lvert a \rvert = \lvert b \rvert \Leftrightarrow a^2 = b^2$ and $\lvert x - a \rvert < b \Leftrightarrow a - b < x < a + b$ in the course of solving equations and inequalities. *e.g. Solve* $\lvert x + 2 \rvert \leq \lvert 2x - 1 \rvert$.	MB7

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1.02m 1.02s 1.02n 1.02t 1.02o 1.02p 1.02q 1.02r	Curve sketching	m) Understand and be able to use graphs of functions. *The difference between plotting and sketching a curve should be known. See Section 2b.* n) Be able to sketch curves defined by simple equations including polynomials. *e.g. Familiarity is expected with sketching a polynomial of degree* ≤ 4 *in factorised form, including repeated roots.* *Sketches may require the determination of stationary points and, where applicable, distinguishing between them.* o) Be able to sketch curves defined by $y=\frac{a}{x}$ and $y=\frac{a}{x^2}$ (including their vertical and horizontal asymptotes). p) Be able to interpret the algebraic solution of equations graphically. q) Be able to use intersection points of graphs to solve equations. *Intersection points may be between two curves one or more of which may be a polynomial, a trigonometric, an exponential or a reciprocal graph.* r) Understand and be able to use proportional relationships and their graphs. *i.e. Understand and use different proportional relationships and relate them to linear, reciprocal or other graphs of variation.*	s) Be able to sketch the graph of the modulus of a linear function involving a single modulus sign. *i.e. Given the graph of* $y=ax+b$ *sketch the graph of* $y=\|ax+b\|$. [*Graphs of the modulus of other functions are excluded.*] t) Be able to solve graphically simple equations and inequalities involving the modulus function.	MB7

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1.02u 1.02v	Functions	*Within Stage 1, learners should understand and be able to apply functions and function notation in an informal sense in the context of the factor theorem (1.02j), transformations of graphs (1.02w), differentiation (Section 1.07) and the Fundamental Theorem of Calculus (1.08a).*	u) Understand and be able to use the definition of a function. *The vocabulary and associated notation is expected i.e. the terms many-one, one-many, one-one, mapping, image, range, domain.* *Includes knowing that a function is a mapping from the domain to the range such that for each x in the domain, there is a unique y in the range with $\mathrm{f}(x) = y$. The range is the set of all possible values of $\mathrm{f}(x)$; learners are expected to use set notation where appropriate.* v) Understand and be able to use inverse functions and their graphs, and composite functions. Know the condition for the inverse function to exist and be able to find the inverse of a function either graphically, by reflection in the line $y = x$, or algebraically. *The vocabulary and associated notation is expected* *e.g.* $\mathrm{gf}(x) = \mathrm{g}(\mathrm{f}(x))$, $\mathrm{f}^2(x)$, $\mathrm{f}^{-1}(x)$.	MB8 OT1.1 OT1.4
1.02w 1.02x	Graph transformations	w) Understand the effect of simple transformations on the graph of $y = \mathrm{f}(x)$ including sketching associated graphs, describing transformations and finding relevant equations: $y = a\mathrm{f}(x)$, $y = \mathrm{f}(x) + a$, $y = \mathrm{f}(x + a)$ and $y = \mathrm{f}(ax)$, for any real a. *Only single transformations will be requested.* *Translations may be specified by a two-dimensional column vector.*	x) Understand the effect of combinations of transformations on the graph of $y = \mathrm{f}(x)$ including sketching associated graphs, describing transformations and finding relevant equations. *The transformations may be combinations of* $y = a\mathrm{f}(x)$, $y = \mathrm{f}(x) + a$, $y = \mathrm{f}(x + a)$ *and* $y = \mathrm{f}(ax)$, *for any real* a, *and* f *any function defined in the Stage 1 or Stage 2 content.*	MB9

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1.02y	Partial fractions		y) Be able to decompose rational functions into partial fractions (denominators not more complicated than squared linear terms and with no more than 3 terms, numerators constant or linear). *i.e. The denominator is no more complicated than* $(ax+b)(cx+d)^2$ *or* $(ax+b)(cx+d)(ex+f)$ *and the numerator is either a constant or linear term.* *Learners should be able to use partial fractions with the binomial expansion to find the power series for an algebraic fraction or as part of solving an integration problem.*	MB10
1.02z	Models in context		z) Be able to use functions in modelling. *Includes consideration of modelling assumptions, limitations and refinements of models, and comparing models.*	MB11

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